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A FINAL REPORT ON

MorphAI: Predicting Facial Appearance Transformation Using Generative Deep Learning

Submitted in partial fulfillment of the requirements for the degree in Bachelor of Computer Engineering

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CERTIFICATE OF APPROVAL

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ABSTRACT

Facial appearance transformation is an important application of generative deep learning that can support facial reconstruction, image editing, and surgical planning. This project aims to develop **MorphAI**, a generative deep learning system for predicting facial appearance transformation while preserving the identity and structural characteristics of the input face. The proposed system uses a modified Local-Global Network (LGNet) with facial landmark conditioning to reconstruct masked or modified facial regions. MediaPipe Face Mesh is used to extract facial landmarks, which are converted into heatmaps and combined with the masked facial image to provide structural information to the model. The system uses reconstruction, perceptual, style, adversarial, and identity-preservation losses to improve the quality and consistency of the generated facial images. An interactive application allows users to upload an image, select facial regions, and generate the reconstructed facial appearance. The developed MorphAI system provides a promising approach for realistic and identity-aware facial reconstruction and can support future applications in facial image editing and computer-assisted surgical planning.

Keywords: *Facial Appearance Transformation, Facial Reconstruction, Generative Adversarial Network, Generative Deep Learning, Identity Preservation, LGNet, MediaPipe Face Mesh.*

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List of Symbols and Abbreviation

Abbreviation	Full Form
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
BOS	Beginning of Sequence
CNN	Convolutional Neural Network
CUDA	Compute Unified Device Architecture
DL	Deep Learning
EOS	End of Sequence
FID	Fréchet Inception Distance
FFN	Feed-Forward Network
GAN	Generative Adversarial Network
GELU	Gaussian Error Linear Unit
GPU	Graphics Processing Unit
IIIT	International Institute of Information Technology
IoU	Intersection over Union
L1	L1 Reconstruction Loss
LaMa	Large Mask Inpainting
LGNet	Large-mask Guided Network
LPIPS	Learned Perceptual Image Patch Similarity
MHA	Multi-Head Attention
ML	Machine Learning
MSE	Mean Squared Error
PSNR	Peak Signal-to-Noise Ratio
RGB	Red, Green, Blue
RMSE	Root Mean Square Error
SSIM	Structural Similarity Index Measure
UI	User Interface
VGG	Visual Geometry Group

CHAPTER 1
INTRODUCTION

1.1 Background Introduction

Facial appearance transformation is an important application of generative deep learning. It can be used for facial image editing, facial reconstruction, and other applications where a modified facial region needs to be generated realistically. Recent advances in deep learning have made it possible to reconstruct missing or masked parts of a face while maintaining the overall appearance and identity of the person.

A face contains many connected features, such as the eyes, eyebrows, nose, and mouth. When one of these regions is removed or modified, generating a realistic replacement requires the model to understand both the local details of the selected region and its relationship with the rest of the face. Generative models such as Generative Adversarial Networks (GANs), autoencoders, and diffusion models have been widely explored for this type of facial reconstruction and editing. PlasticGAN explored facial changes related to plastic and aesthetic surgery, while Mask-aware Photorealistic Facial Attribute Manipulation used facial masks to control changes in specific facial regions [1, 2].

Several studies have also explored deep learning methods for facial editing and reconstruction. TransEditor used a Transformer-based GAN for controllable facial editing, while Stitch it in Time used a StyleGAN-based approach for editing real facial videos [3,4]. These studies show that generative models can produce realistic facial changes, but many existing approaches are designed for general facial editing and may require complex processing or high computational resources.

Facial reconstruction has also been studied for applications related to surgical planning. Fang et al. proposed a deep learning method to predict facial appearance changes from planned changes in facial bone structure [5]. Bini et al. combined image inpainting and depth estimation for facial reconstruction and surgical outcome prediction [6]. Similarly, Xu et al. developed a diffusion-based method for restoring facial regions while preserving identity [7]. These studies show the importance of maintaining facial structure and identity when reconstructing facial images.

Facial landmarks can provide useful structural information for facial reconstruction. DeepFaceReshaping demonstrated the use of facial landmarks to control facial reshaping [8]. In MorphAI, facial landmarks are extracted using MediaPipe Face Mesh and are used to generate a heatmap that provides additional facial structure information to the model. The system supports reconstruction of selected facial regions, including the left eye, right eye, left eyebrow, right eyebrow, nose, and mouth. It also supports different mask sizes for testing the reconstruction process.

Based on these ideas, this project proposes **MorphAI: Predicting Facial Appearance Transformation Using Generative Deep Learning**. The system uses a modified Local-Global Network (LGNet) as the generator to reconstruct masked facial regions. The model receives the masked image together with the mask and facial landmark heatmap as input. During training, a PatchGAN-based discriminator is used with the generator to improve the realism of the reconstructed images. The training process uses reconstruction, perceptual, style, identity, and facial landmark losses to guide the generation process.

The developed MorphAI application allows a user to upload a facial image, select a facial region or mask type, and generate a reconstructed result. The system provides the original image, masked image, and reconstructed image for comparison. It also calculates image quality and identity-related metrics during evaluation.

1.2 Motivation

Facial appearance reconstruction is challenging because a realistic result must preserve the important features and identity of the input face while generating the missing or masked regions naturally. Existing facial editing and reconstruction methods use different approaches, including GANs, diffusion models, and landmark-based methods, but many are computationally expensive or are designed for general facial editing rather than controlled facial region reconstruction. These limitations motivate the development of a system that can use facial structural information to guide the reconstruction process. MorphAI addresses this problem by combining facial landmarks extracted using MediaPipe Face Mesh with a modified LGNet-based generative model to reconstruct selected facial regions. The use of landmark information, adversarial learning, and identity-related constraints aims to produce realistic facial reconstructions while maintaining the overall appearance of the input face.

1.3 Problem Statement

Facial appearance reconstruction in MorphAI involves several key challenges:

- **Realistic reconstruction:** The missing or masked facial region should look natural and match the surrounding face.
- **Identity preservation:** The reconstruction should preserve the important facial characteristics of the original person.
- **Facial structure:** Facial features have fixed spatial relationships, so the generated region should be correctly aligned with the rest of the face.
- **Different facial regions:** The system needs to reconstruct different regions such as the eyes, eyebrows, nose, and mouth while maintaining a consistent facial appearance.
- **Guidance from landmarks:** The masked image alone may not provide enough structural information. Facial landmarks can help the model understand the position and shape of important facial features.
- **Realistic image generation:** The generated result should have clear details and appear visually similar to a real face. Adversarial learning helps improve the realism of the reconstructed image.
- **Limitations of existing methods:** Many existing facial editing and reconstruction methods require complex processing or high computational resources. MorphAI aims to provide a practical approach using a modified LGNet with facial landmark information and generative learning [3, 4, 7].

1.4 Objective

The main objective of this project is:

- To develop **MorphAI**, a generative deep learning system for predicting facial appearance transformation using facial landmarks to guide reconstruction while preserving the identity and structure of the input face.

Scope Clarification: Reconstruction vs. Surgical Prediction

It is important to clarify the distinction between facial image reconstruction and clinical surgical outcome prediction. MorphAI is a deep learning-based facial image inpainting and reconstruction framework: it reconstructs synthetically masked facial regions (such as the eyes, eyebrows, nose, and mouth) using surrounding facial context and landmark-based structural information. The system does not simulate biological tissue deformation, healing processes, anatomical changes, or actual post-operative surgical outcomes. Since the experiments use the CelebA-HQ dataset with synthetically applied masks rather than real clinical pre-operative and post-operative patient data, MorphAI's output should be interpreted as a facial reconstruction result and not as a prediction of how a person's face would look after surgery.

1.5 Scope of Project

The scope of this project includes:

- Developing a generative deep learning system for facial region reconstruction (inpainting) using synthetically masked facial images.
- Reconstructing selected facial regions such as the eyes, eyebrows, nose, and mouth from a masked facial image using landmark-conditioned generation.
- Extracting facial landmarks using MediaPipe Face Mesh to provide structural guidance for the reconstruction process.
- Evaluating reconstruction quality and identity preservation using standard image-quality and face-recognition metrics.

Potential Future Applications

While MorphAI currently focuses on facial region reconstruction, the approach could be extended to other areas in the future. These are possible directions, not features that have been built or tested:

- **Facial Image Editing and Restoration:** The same reconstruction approach could be used to fix damaged or blocked parts of facial images in general, not just the regions tested in this project.
- **CCTV/Video Restoration:** With further work, this method could help reconstruct faces that are partly blocked or hidden in low-quality surveillance footage.

- **AR/VR and Gaming:** The landmark-based approach could be adapted for building or completing facial avatars in AR/VR and gaming applications.
- **Surgical Visualization:** This could be a starting point for future research into facial visualization tools. However, MorphAI has not been trained or tested on real clinical data, so it is not ready for actual surgical planning or predicting surgical outcomes. That would require real before-and-after patient data and proper clinical testing, which is beyond the scope of this project.

CHAPTER 2
LITERATURE REVIEW

2.1 Literature Review

Facial appearance transformation has developed from traditional image editing to deep learning-based methods such as GANs, Transformers, and diffusion models. These methods can reconstruct or modify facial regions, but maintaining the person's identity and natural facial structure is still challenging.

GAN-based methods such as PlasticGAN and mask-aware facial manipulation have shown good results in facial editing [1,2]. Other methods, including TransEditor and Stitch it in Time, have improved controllable facial editing [3, 4]. However, many existing methods require high computational resources or are designed mainly for general facial editing.

Recent studies have also explored facial reconstruction for surgical planning and appearance prediction [5, 6]. Diffusion and landmark-based methods have further improved facial restoration and structural control [7, 8]. These studies provide the basis for MorphAI, which uses MediaPipe Face Mesh landmarks and a modified LGNet to reconstruct selected facial regions while preserving facial structure and identity.

2.1.1 Diffuse and Restore: A Region-Adaptive Diffusion Model for Identity-Preserving Blind Face Restoration (2024)

Xu et al. (2024) proposed a diffusion-based method for restoring degraded facial images while preserving identity. The model uses an Identity Preserving Conditioner (IPC) with a Conditional Diffusion Probabilistic Model (DDPM) to restore facial regions. However, it focuses mainly on face restoration rather than facial appearance transformation or surgical prediction [7].

2.1.2 PlasticGAN: Holistic Generative Adversarial Network on Face Plastic and Aesthetic Surgery (2022)

Chandaliya and Nain (2022) proposed PlasticGAN for generating facial appearance changes related to plastic and aesthetic surgery. The model uses a GAN with age and gender information to generate realistic facial transformations. However, the study uses relatively low-resolution images and a limited evaluation setup [1].

2.1.3 TransEditor: Transformer-Based Dual-Space GAN for Highly Controllable Facial Editing (2022)

Xu et al. (2022) introduced TransEditor, which combines a Transformer with StyleGAN2 to provide controlled facial editing. It separates facial structure and appearance into different latent spaces to achieve realistic modifications. However, the method requires latent inversion and considerable computational resources [3].

2.1.4 Stitch it in Time: GAN-Based Facial Editing of Real Videos (2022)

Tzaban et al. (2022) proposed a StyleGAN2-based method for editing facial attributes in real videos while maintaining identity and temporal consistency. The method produces smooth facial changes without noticeable flickering. However, it is mainly designed for video editing rather than facial reconstruction or surgical appearance prediction [4].

2.1.5 Deep Learning Framework for Facial Reconstruction Outcome Prediction (2025)

Bini et al. (2025) proposed a deep learning framework for facial reconstruction and computer-assisted surgical planning. The study combines image inpainting and depth estimation and evaluates different models for facial reconstruction. However, the approach does not specifically focus on identity preservation, which is an important aspect of the proposed MorphAI system [6].

2.1.6 DeepFaceReshaping: Interactive Deep Face Reshaping via Landmark Manipulation (2024)

Chen et al. (2024) proposed an interactive facial reshaping method based on facial landmark manipulation. The framework combines Graph VAE, Conditional GAN, Transformer, and U-Net to generate realistic facial changes. However, it mainly focuses on 2D facial editing and is not specifically designed for surgical outcome prediction [8].

2.1.7 Deep Learning-based Facial Appearance Simulation Driven by Surgically Planned Craniomaxillofacial Bony Movement (2022)

Fang et al. (2022) proposed a deep learning method for simulating facial appearance changes based on surgically planned craniomaxillofacial bone movement. The approach learns the relationship between structural changes and facial appearance. However, it requires specialized patient-specific anatomical data and focuses mainly on craniomaxillofacial surgical simulation [5].

2.1.8 Mask-aware Photo Facial Attribute Manipulation (2021)

Sun et al. (2021) proposed a mask-aware method for controlling facial attribute manipulation. The approach uses masked facial regions to guide the generation process while maintaining realistic facial details. However, it focuses on facial attribute editing rather than facial reconstruction for surgical prediction, and additional constraints are needed to preserve identity and facial details [2].

2.1.9 Research Gap

Previous studies have shown good results in facial restoration, editing, and surgical appearance prediction. However, most methods focus on general face editing, image restoration, or complex surgical simulation rather than localized facial transformation.

Another issue is identity preservation. A generated face may look realistic but still change the person's identity or facial structure. Some methods also require large computational resources or special 3D patient data, which may not be practical for a simple image-based system.

Therefore, there is a need for a simpler framework that can transform a selected facial region while keeping the person's identity and facial structure. MorphAI addresses this gap by combining MediaPipe Face Mesh landmarks with a modified LGNet model and ArcFace-based identity loss. This allows the system to use facial structure as guidance and generate realistic facial reconstruction using only 2D facial images.

2.2 Related Theories

2.2.1 Facial Image Reconstruction

Facial image reconstruction aims to recover missing or corrupted regions of a face using the surrounding information. In MorphAI, this is treated as a guided facial inpainting problem where a mask indicates the region to be reconstructed.

The masked image is obtained as:

$$I_{\text{masked}} = I \odot (1 - M) \tag{2.1}$$

where I is the original image, M is the binary mask, and $\odot$ denotes element-wise multiplication.

2.2.2 Facial Landmark Detection

Facial landmark detection identifies important points such as the eyes, eyebrows, nose, and mouth. MorphAI uses these points to provide facial structure information and to generate semantic masks and landmark heatmaps.

2.2.3 MediaPipe Face Mesh

MediaPipe Face Mesh is used in MorphAI to extract facial landmarks from input images. The detected landmarks are grouped into facial regions and used for mask generation and structural guidance.

2.2.4 Facial Region Masking

Facial masking identifies the region that needs to be reconstructed. MorphAI uses both semantic facial masks and random rectangular masks. Semantic masks are generated from facial landmarks and can represent regions such as the eyes, eyebrows, nose, and mouth.

2.2.5 Generative Adversarial Networks

Generative Adversarial Networks (GANs) consist of a generator and a discriminator. The generator produces reconstructed images, while the discriminator evaluates whether the generated results appear realistic.
In MorphAI, adversarial learning helps improve the visual quality and realism of reconstructed facial regions.

2.2.6 Generator and Discriminator

The generator reconstructs the masked facial region using the available image information, mask, and landmark guidance. The discriminator evaluates the generated image against real facial images.
Together, they encourage the generator to produce realistic facial details and textures.

2.2.7 PatchGAN

PatchGAN evaluates local image patches instead of judging the complete image with a single value. This allows the discriminator to focus on details such as skin texture, eyes, lips, and facial boundaries.
MorphAI uses a PatchGAN discriminator to improve the local realism of reconstructed facial regions.

2.2.8 Local-Global Network (LGNet)

LGNet uses local and global information to reconstruct missing image regions. The local pathway focuses on details inside the missing region, while the global pathway uses surrounding information to maintain overall consistency.
MorphAI uses a modified LGNet generator with landmark heatmap guidance. The generator receives the masked image, facial mask, and landmark heatmap:

$$\hat{I}_{\text{raw}} = G(I_{\text{masked}}, M, H) \tag{2.2}$$

where G represents the LGNet generator and H represents the landmark heatmap.

2.2.9 Loss Functions

MorphAI uses multiple loss components to guide facial reconstruction and preserve visual quality, texture, identity, and facial structure. The main loss components are:

- **L1 Loss:** Measures the pixel-level difference between the reconstructed image and the target image. It helps the model produce accurate pixel-level reconstruction.

$$\mathcal{L}_{L1} = \frac{1}{CHW} \sum_{c,h,w} \left| \hat{I}_{c,h,w} - I_{c,h,w} \right| \tag{2.3}$$

- **Perceptual Loss:** Compares deep feature representations of the reconstructed and target images. It helps preserve important facial structures and visual details.

$$\mathcal{L}_{perceptual} = \sum_{j} \frac{1}{C_j H_j W_j} \left| \phi_j(\hat{I}) - \phi_j(I) \right|_1 \tag{2.4}$$

- **Style Loss:** Compares feature correlations between the reconstructed and target images to maintain similar texture and appearance.

$$\mathcal{L}_{style} = \sum_j \frac{1}{C_j^2} \left| G_j(\hat{I}) - G_j(I) \right|_1 \tag{2.5}$$

- **Identity Loss:** Encourages the reconstructed face to preserve the identity of the input subject using facial feature embeddings and cosine similarity.

$$\mathcal{L}_{identity} = 1 - \frac{f(\hat{I}) \cdot f(I)}{\|f(\hat{I})\|_2 \|f(I)\|_2} \tag{2.6}$$

- **Facial Landmark Loss:** Encourages the reconstructed image to maintain the correct position of important facial features.

$$\mathcal{L}_{landmark} = \frac{1}{N} \sum_{k=1}^{N} \text{Wing} \left(\text{Det}(\hat{I})_k - \text{Det}(I)_k \right) \tag{2.7}$$

- **Adversarial Loss:** Encourages the generator to produce reconstructions that the PatchGAN discriminator cannot distinguish from real facial images, improving perceptual realism and local texture detail.

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_I \left[(D(I) - 1)^2 \right] + \mathbb{E}_{\hat{I}} \left[D(\hat{I})^2 \right] \tag{2.8}$$

  where D denotes the PatchGAN discriminator, I the ground-truth facial image, and $\hat{I}$ the reconstructed image generated by LGNet.

2.2.10 Identity Preservation

Identity preservation ensures that the reconstructed face remains visually consistent with the input subject. MorphAI uses facial embeddings and ArcFace-based similarity during evaluation to measure identity preservation.

2.2.11 Boundary Blending and Post-processing

The reconstructed region may contain visible boundaries between the generated and original areas. MorphAI therefore applies blending during post-processing to create a smoother transition between the reconstructed region and the surrounding face. For alpha feathering:

$$\alpha = M * G_\sigma \tag{2.9}$$

where G_σ represents a Gaussian kernel. The final blended image is given by:

$$I_{\text{final}} = \alpha \hat{I} + (1 - \alpha) I_{\text{bg}} \tag{2.10}$$

where $\hat{I}$ is the reconstructed image and I_{bg} is the original background.
In addition to alpha feathering, MorphAI also supports **seamless Poisson cloning** as a blending option. Instead of blending pixel values directly, Poisson blending matches the gradients of the reconstructed region to the surrounding image, producing smoother colour and illumination consistency across the boundary at a higher computational cost than alpha feathering. Both options are available to the user in the MorphAI interface, allowing a trade-off between blending speed and visual smoothness.

2.2.12 Evaluation Metrics

MorphAI evaluates facial reconstruction using pixel-level, structural, perceptual, realism, and identity-based metrics:

- **Peak Signal-to-Noise Ratio (PSNR):** Measures pixel-level similarity between the reconstructed and target images. Higher values indicate better reconstruction quality.
- **Structural Similarity Index Measure (SSIM):** Measures structural similarity between the reconstructed and target images. Higher values indicate better preservation of facial structure.
- **Learned Perceptual Image Patch Similarity (LPIPS):** Measures perceptual similarity using deep image features. Lower values indicate greater perceptual similarity.
- **Fréchet Inception Distance (FID):** Compares the feature distributions of reconstructed and real images. Lower values generally indicate better visual realism.
- **Identity Similarity:** Measures how well the reconstructed face preserves the identity of the input subject using facial embeddings and cosine similarity. Higher values indicate better identity preservation. ArcFace is used for identity evaluation.
- **Root Mean Square Error (RMSE):** Measures the pixel-level reconstruction error between the reconstructed and target images. Lower values indicate smaller reconstruction errors.
- **RMSE-log:** Measures reconstruction error after applying a logarithmic transformation to the reconstructed and ground-truth images, reducing the

influence of large pixel differences. Lower values indicate smaller reconstruction errors. The full mathematical formulation is given in Section 3.5.7 (Eq. 3.8).

2.2.13 Dataset

2.2.13.1 Dataset Description

MorphAI uses the CelebA-HQ dataset for facial image reconstruction. The dataset contains high-quality facial images with diverse identities, expressions, poses, and appearances. The images are used as ground-truth facial images for training and evaluation.
During training, facial regions are artificially masked so that the model learns to reconstruct the missing areas. The images are resized to 256×256 pixels to match the input requirements of the LGNet generator.
It is important to note that CelebA-HQ contains only single, unpaired facial images – it does not provide real before-and-after surgical image pairs. The facial reconstruction task used to train and evaluate MorphAI is created artificially: selected facial regions are synthetically masked on an otherwise complete image, and the original unmasked image serves as the ground truth for reconstruction. No real pre-operative or post-operative surgical patient dataset is used anywhere in this project. This synthetic masking approach approximates a region-completion task; it does not model or represent actual surgical tissue change.

2.2.13.2 Dataset Preprocessing

The facial images are resized to 256×256 pixels and processed before being provided to the model. Facial landmarks are extracted using MediaPipe Face Mesh and are used to provide structural information about the face.
The RGB image, binary facial mask, and landmark heatmap are combined to form a five-channel input:

$$3 \text{ RGB channels} + 1 \text{ mask channel} + 1 \text{ heatmap channel} = 5 \text{ channels}. \quad (2.11)$$

Therefore, the input to the LGNet generator has the dimensions $5 \times 256 \times 256$.

2.2.13.3 Dataset Split

The CelebA-HQ dataset is divided into training, validation, and test sets. The training set is used to optimize the model, the validation set is used to monitor

training performance, and the test set is used for final evaluation.

2.2.13.4 Mask Generation

MorphAI uses facial region masks to simulate missing parts of the face. The masks are generated for different facial regions, including:

- **Eyes and eyebrows**
- **Nose**
- **Mouth**

These masks allow the model to learn how to reconstruct different parts of the face from the available surrounding information.

2.2.13.5 Data Representation

Each training sample consists of an RGB facial image, a binary mask, and a landmark heatmap. These components are combined as:

$$X = [I_{\text{RGB}}, M, H] \tag{2.12}$$

where I_{RGB} represents the facial image, M represents the facial mask, and H represents the landmark heatmap.

2.2.13.6 Dataset Challenges

The main challenges associated with using CelebA for MorphAI include:

- **Initially:** During facial alignment of the CelebA images, inaccurate landmark detection occasionally caused improper alignment and cropping, particularly around the chin region.
- **Image Alignment:** Facial images require proper alignment to maintain consistent facial structure and improve the reconstruction process.
- **Image Resizing:** Images must be resized to the required 256×256 resolution while preserving important facial details.
- **Synthetic Masking:** The dataset does not provide actual before-and-after facial surgery pairs, so missing regions are created artificially.
- **Facial Variation:** Differences in pose, expression, and appearance can make facial reconstruction more challenging.

- **Lighting Variation:** Changes in illumination may affect the appearance and reconstruction of facial regions.
- **Landmark Accuracy:** Incorrect facial landmarks can affect the generated landmark heatmap and reconstruction quality.

2.3 Review Matrix

Table 2.1 presents a summary of the literature reviewed for the MorphAI project, highlighting key methodologies, datasets, and limitations in facial appearance transformation and reconstruction.

Table 2.1: Summary of Reviewed Literature for MorphAI

S.N.	**Author(s) & Year**	**Method**	**Dataset**	**Limitations**
1	Xu et al. (2024) [7] *Diffuse and Restore*	Identity Preserving Conditioner (IPC) + Conditional DDPM	FFHQ, CelebA-Test	Designed for face restoration rather than surgical prediction; diffusion-based models require high computational resources.
2	Chandaliya and Nain (2022) [1] *PlasticGAN: Holistic Generative Adversarial Network on Face Plastic and Aesthetic Surgery*	GAN with age and gender conditioning	CACD, UTKFace, CLF, Adience	Uses a small evaluation dataset and relatively low image resolution (128×128); based on an older GAN architecture.
3	Xu et al. (2022) [3] *TransEditor*	StyleGAN2 + Transformer + Dual Latent Space	FFHQ, CelebA-HQ	Requires latent inversion and high GPU memory; performance can be affected by large pose variations.

Continued on next page

Table 2.1 (continued)

S.N.	Author(s) & Year	Method	Dataset	Limitations
4	Tzaban et al. (2022) [4] *Stitch it in Time*	StyleGAN2-based GAN	FFHQ	Designed primarily for facial video editing rather than surgical prediction and requires considerable computational resources.
5	Bini et al. (2025) [6] *AI-Based Facial Reconstruction for Surgical Outcome Prediction*	LGNet + DepthPro	FFHQ, C3I-SynFace	Uses benchmark datasets instead of real patient data; healing processes and tissue deformation are not modeled.
6	Chen et al. (2024) [8] *DeepFaceReshaping*	Graph VAE + Conditional GAN + Transformer + U-Net	CelebA-HQ	Limited to 2D facial editing and mainly frontal faces; not specifically designed for clinical surgical prediction.
7	Fang et al. (2022) [5] *Deep Learning-based Facial Appearance Simulation Driven by Surgically Planned Craniomaxillofacial Bony Movement*	Attentive Correspondence assisted Movement Transformation Network (ACMT-Net)	Patient-specific facial and bony structure data	Focuses on facial appearance prediction driven by surgically planned bony movement; requires specialized anatomical data and is mainly intended for craniomaxillofacial surgical simulation.

Continued on next page

Table 2.1 (continued)

S.N.	Author(s) & Year	Method	Dataset	Limitations
8	Sun et al. (2021) [2] *Mask-aware Photorealistic Facial Attribute Manipulation*	Mask-Adversarial Autoencoder (M-AAE): VAE + GAN	Facial attribute manipulation datasets	Focuses on facial attribute manipulation rather than surgical prediction; additional supervision is required to maintain identity and facial details.

CHAPTER 3

METHODOLOGY

3.1 MorphAI System Workflow Diagram

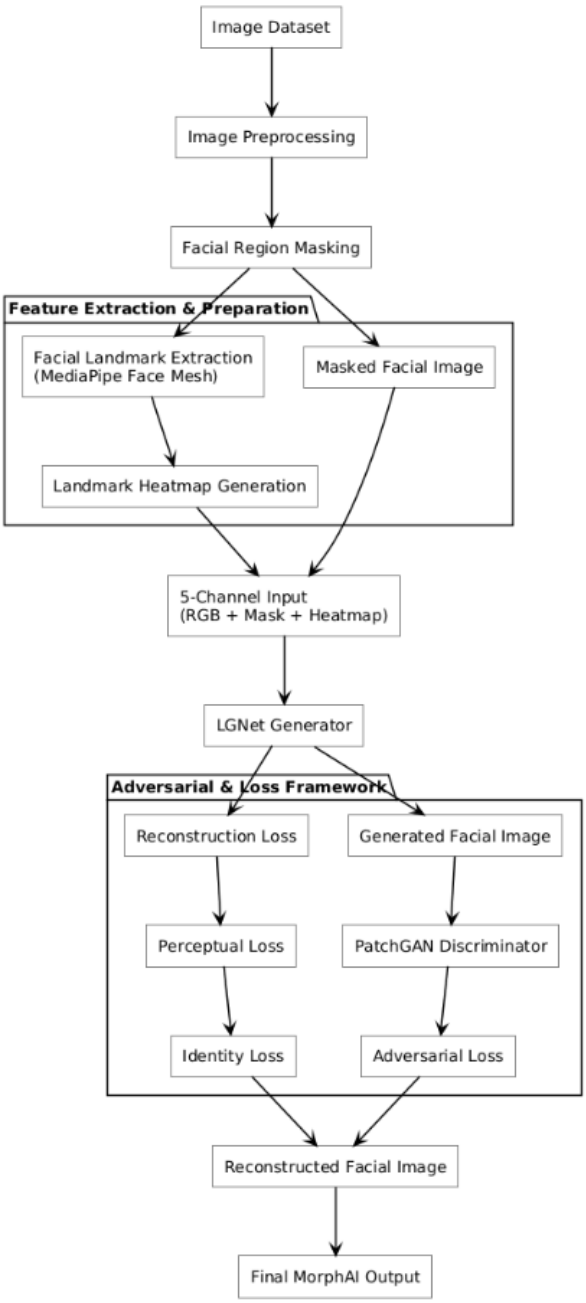


Figure 3.1: Stages of MorphAI System

3.2 Description of Working Flow of the System

The working process of MorphAI consists of several stages, starting from the input facial image and ending with the reconstructed facial image. The main stages of the system are described below.

3.2.1 Image Dataset

MorphAI uses facial images as the input dataset for training and evaluation. The original facial images are treated as ground-truth images. During training, different facial regions are masked so that the model learns to reconstruct the missing regions.

3.2.2 Image Preprocessing

The input images are resized and converted into a suitable format for the model. Facial landmark detection is then performed on the processed images to obtain the structural information required by the reconstruction system.

3.2.3 Facial Region Masking

A facial region is selected for reconstruction using a binary mask. The mask identifies the area that will be removed or hidden from the input image. MorphAI supports facial regions such as the eyes, eyebrows, nose, and mouth.

3.2.4 Feature Extraction and Preparation

After masking, the system prepares the information required by the reconstruction model. MediaPipe Face Mesh is used to extract facial landmarks from the input image.

The detected landmarks are then used to generate a landmark heatmap. The heatmap provides additional information about the position and structure of facial features.

At the same time, the masked facial image is prepared as a separate input. These components are then combined to form the model input.

3.2.5 Five-Channel Input Formation

MorphAI combines the RGB image, facial mask, and landmark heatmap to create a five-channel input:

$$3 \text{ RGB channels} + 1 \text{ mask channel} + 1 \text{ heatmap channel} = 5 \text{ channels}. \quad (3.1)$$

The resulting input has the form:

$$X \in \mathbb{R}^{5\times256\times256}. \quad (3.2)$$

This allows the generator to use both visual information and facial structural information during reconstruction.

3.2.6 LGNet Generator

The five-channel input is passed to the modified LGNet generator. The generator uses the visible facial information, mask, and landmark heatmap to reconstruct the missing facial region.
The output of the generator is a generated facial image containing the reconstructed region.

3.2.6.1 Loss Framework

The generated image is optimized using a composite loss framework during training. The loss function combines reconstruction, perceptual, style, identity, facial landmark, and adversarial losses to guide the generator toward accurate reconstruction, preservation of facial structure, texture consistency, identity preservation, and visual realism. The individual loss terms are defined, and the total training objective is given by:

$$\mathcal{L}_{\text{total}} = \lambda_1\mathcal{L}_{\text{L1}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perceptual}} + \lambda_{\text{style}}\mathcal{L}_{\text{style}} + \lambda_{\text{id}}\mathcal{L}_{\text{identity}} + \lambda_{\text{lm}}\mathcal{L}_{\text{landmark}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} \quad (3.3)$$

where λ_1, λ_{perc}, λ_{style}, λ_{id}, λ_{lm}, and λ_{adv} are the weighting coefficients for each loss term.

3.2.7 Reconstructed Facial Image

After the generator and loss-based optimization, the reconstructed facial image is obtained. The generated region is combined with the remaining facial region to produce a complete reconstructed face.

3.2.7.1 Final MorphAI Output

The final output of MorphAI is the reconstructed facial image. The output is intended to preserve the facial structure, appearance, and identity of the input image while reconstructing the selected facial region.

3.3 Model Architecture

MorphAI adopts a facial inpainting architecture designed to reconstruct selected facial regions while preserving the overall appearance and identity of the input face. The system combines facial landmark extraction, mask generation, landmark-based guidance, and the LGNet generator to produce the reconstructed image.

3.3.1 Input Image

The system receives a facial image from the user through the frontend. The image is prepared and passed to the backend for facial region selection and reconstruction.

3.3.2 Facial Landmark Extraction

The LandmarkExtractor detects important facial landmarks from the input image. These landmarks provide structural information about the face and help guide the reconstruction of the selected region.

3.3.3 Mask Generation

The MaskGenerator creates a region-specific binary mask based on the selected facial area. The mask identifies the region to be reconstructed while preserving the remaining facial content.

3.3.4 Landmark Heatmap

The detected landmarks are converted into a spatial heatmap. This heatmap provides the generator with additional information about the facial structure and location of important facial features.

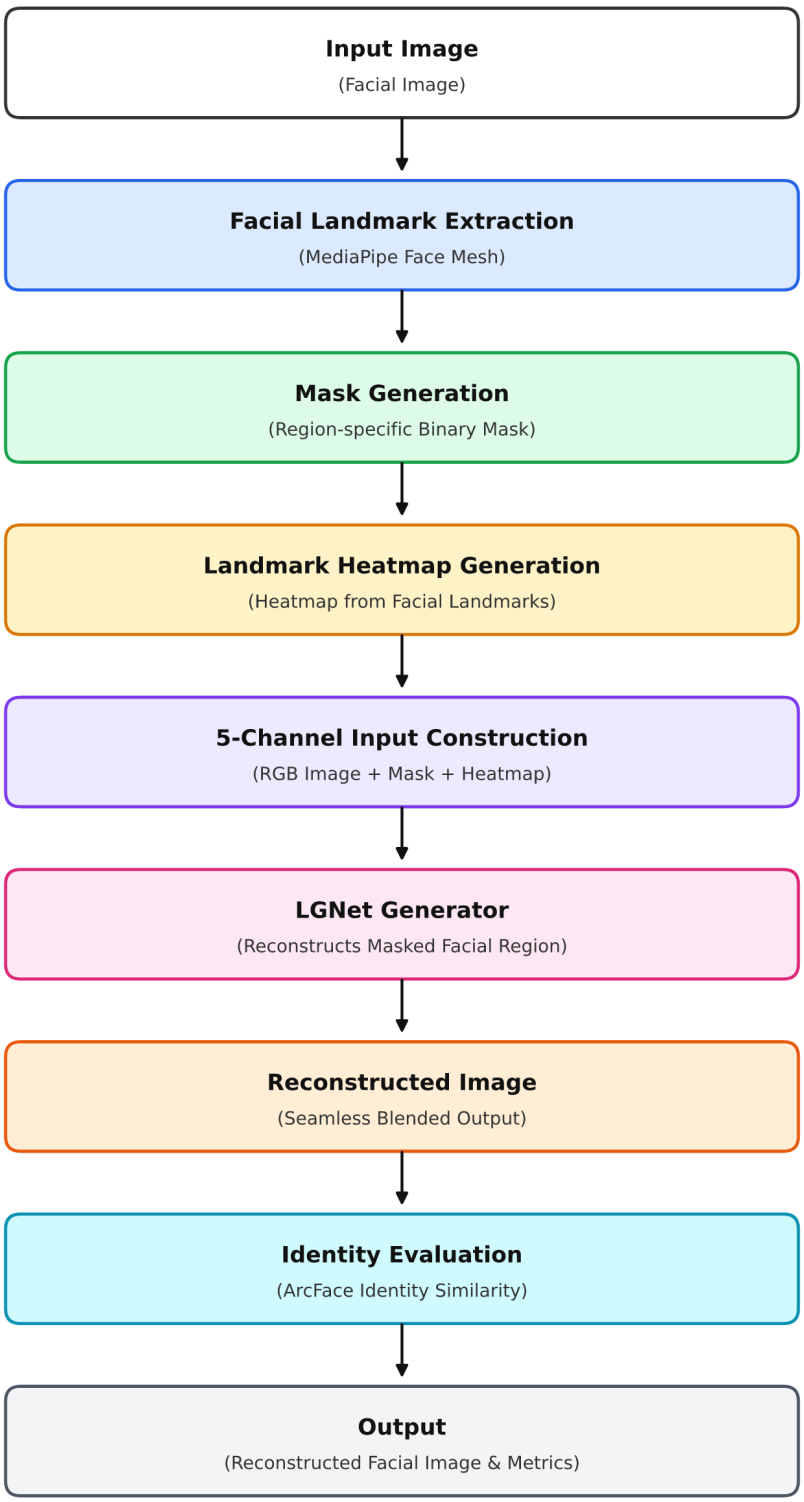


Figure 3.2: MorphAI model architecture

3.3.5 LGNet Generator

The LGNetGenerator is the main reconstruction component of MorphAI. It receives the RGB image, facial mask, and landmark-based information as the required multi-channel input. The generator processes these inputs and reconstructs the selected facial region while maintaining consistency with the surrounding facial features.

3.3.6 Identity Evaluation

The reconstructed image can be evaluated using the IdentityExtractor to determine whether the generated face maintains the identity characteristics of the original input. This provides an additional measure for assessing the quality of the reconstruction.

3.3.7 Model Architecture Summary

Table 3.1 summarises the main components of the MorphAI architecture.

Table 3.1: MorphAI model architecture summary

Component	**Purpose**
Frontend	Image upload and region selection
LandmarkExtractor	Detect facial landmarks
MaskGenerator	Generate the reconstruction mask
Landmark Heatmap	Provide facial structural guidance
LGNetGenerator	Reconstruct the selected facial region
IdentityExtractor	Evaluate identity preservation

3.3.8 Technical Contribution of MorphAI

The primary technical contribution of MorphAI is the incorporation of explicit facial landmark heatmap conditioning into a modified LGNet-based facial inpainting framework. Rather than introducing an entirely new generative architecture, MorphAI builds on the existing Local-Global Network (LGNet) generator and extends it with structural guidance derived from facial landmarks.
The model receives a five-channel input consisting of the RGB facial image (3 channels), a binary facial mask (1 channel), and a landmark heatmap (1 channel):

$$X = [I_{RGB}, M, H] \in \mathbb{R}^{5\times256\times256} \tag{3.4}$$

The landmark heatmap H provides the generator with explicit spatial information about the position and shape of important facial features, which is not available to the original LGNet or to context-only inpainting models such as LaMa. This allows the generator to combine two complementary sources of information during reconstruction: the visual context surrounding the masked region, and the underlying facial geometry indicated by the landmarks.
In addition to landmark conditioning, MorphAI incorporates identity-preserving supervision using ArcFace-based facial embeddings during training, encouraging the reconstructed region to remain consistent with the identity of the input subject. This distinguishes MorphAI from landmark-based methods such asDeepFaceReshaping [8], which use landmarks primarily to drive interactive geometric reshaping of *visible* facial regions, rather than as a conditioning signal for reconstructing *missing* regions.
Overall, the novelty of MorphAI lies in the combination of the following elements, rather than in proposing a new generative architecture:

1. Landmark heatmap conditioning,
2. Identity-preserving loss supervision, and
3. Application of this combined approach to region-specific facial inpainting.

MorphAI is therefore best described as a landmark-conditioned, identity-preserving adaptation of LGNet for facial region reconstruction.

3.4 Hyperparameters

The MorphAI model was trained using PyTorch with Automatic Mixed Precision (autocast). The training configuration was selected to balance reconstruction quality, facial structure, texture consistency, and identity preservation.

- **Training Configuration:** The MorphAI model was trained for 5 epochs using the CelebA-HQ dataset containing 16,800 facial images. A portion of these images was held out for evaluation.
- **Loss Function:** A composite loss combining L1, perceptual, style, identity, and facial landmark losses was used to improve reconstruction quality and preserve facial characteristics.
- **L1 Loss:** Measures pixel-level differences between the reconstructed and target images.

- **Perceptual Loss:** Uses VGG-16 features to preserve important facial structures and visual details.
- **Style Loss:** Uses Gram matrix features to maintain texture and appearance consistency.
- **Identity Loss:** Uses facial embedding cosine similarity to preserve the identity of the input subject.
- **Landmark Loss:** Uses Wing loss to maintain the position and structure of important facial landmarks.
- **Adversarial Loss:** Uses a PatchGAN discriminator trained jointly with the generator to improve the local realism and texture fidelity of reconstructed facial regions.
- **Training Efficiency:** Automatic Mixed Precision was used to reduce GPU memory consumption and improve training efficiency.
- **Training Convergence:** Training loss decreased from 0.3479 in the first epoch to 0.2774 in the fifth epoch, while validation loss decreased from 0.3640 to 0.2950, indicating stable convergence.
- **Final Checkpoint:** The fifth-epoch checkpoint, `morphai_epoch_05.pt`, was selected as the final model for evaluation and comparison with the LaMa baseline.

Table 3.2: Training configuration of the proposed LGNet model

Training Setting	**Value**
Framework	PyTorch
Training method	Automatic Mixed Precision (autocast)
Training epochs	5
Dataset	CelebA-HQ (16,800 facial images)
Loss function	Composite multi-objective loss
Pixel loss	L1 loss
Perceptual loss	VGG-16 based
Style loss	Gram matrix based
Adversarial loss PatchGAN discriminator	
Identity loss	Face embedding cosine loss
Landmark loss	Wing loss
Final checkpoint	`morphai_epoch_05.pt`

3.5 Performance Evaluation Metrics

MorphAI is evaluated using metrics that measure reconstruction quality, visual similarity, and identity preservation. The evaluation was performed using 2,800 held-out facial images. For each held-out image, six facial-region masks were generated for the left eyebrow, right eyebrow, left eye, right eye, nose, and mouth. Therefore, the evaluation consisted of $2,800 \times 6 = 16,800$ region-specific evaluation samples. These 16,800 samples are derived evaluation instances and do not represent 16,800 unique facial images.

3.5.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the pixel-level similarity between the reconstructed image and the original image. A higher PSNR value indicates that the reconstructed image is closer to the ground-truth image.

$$\text{PSNR} = 10 \log_{10} \left(\frac{MAX_I^2}{\text{MSE}} \right) \tag{3.5}$$

where MAX_I is the maximum possible pixel value and MSE is the mean squared error.

3.5.2 Structural Similarity Index (SSIM)

SSIM measures how similar the structure of the reconstructed image is to the original image. It considers factors such as brightness, contrast, and structural information. Higher SSIM indicates better structural similarity.

3.5.3 Learned Perceptual Image Patch Similarity (LPIPS)

LPIPS measures perceptual similarity using features extracted from a deep neural network. It is useful for checking whether the reconstructed facial region looks visually similar to the original. Lower LPIPS values indicate better perceptual similarity.

3.5.4 Fréchet Inception Distance (FID)

FID compares the feature distributions of generated and real images. It is mainly used to measure the realism of generated images. A lower FID value indicates that the generated images are closer to real facial images.

3.5.5 Identity Similarity

Identity similarity measures whether the reconstructed face still represents the same person as the original image. MorphAI uses facial embeddings and cosine similarity for this evaluation.

$$\text{Identity Similarity} = \frac{f(\hat{I}) \cdot f(I)}{\|f(\hat{I})\|_2 \|f(I)\|_2} \tag{3.6}$$

where $f(I)$ and $f(\hat{I})$ represent the facial embeddings of the original and reconstructed images, respectively. Higher similarity indicates better identity preservation.

3.5.6 Root Mean Square Error (RMSE)

RMSE measures the average pixel-level difference between the reconstructed image and the ground-truth image. Lower RMSE indicates smaller reconstruction errors.

$$\text{RMSE} = \sqrt{\frac{1}{CHW} \sum_{c=1}^{C} \sum_{h=1}^{H} \sum_{w=1}^{W} \left(\hat{I}_{c,h,w} - I_{c,h,w} \right)^2} \tag{3.7}$$

where $\hat{I}$ is the reconstructed image and I is the ground-truth image.

3.5.7 RMSE-log

RMSE-log measures the reconstruction error after applying a logarithmic transformation to the pixel values. It reduces the influence of large pixel differences and provides an error measure on a logarithmic scale.
The RMSE-log is defined as:

$$RMSE_{\log} = \sqrt{\frac{1}{CHW} \sum_{c=1}^{C} \sum_{h=1}^{H} \sum_{w=1}^{W} \left[\log(1 + \hat{I}_{c,h,w}) - \log(1 + I_{c,h,w}) \right]^2} \tag{3.8}$$

where I represents the ground-truth image, $\hat{I}$ represents the reconstructed image, C is the number of image channels, and H and W are the image height and width, respectively. The $\log(1+x)$ transformation allows zero-valued pixels to be handled safely.
Lower RMSE-log values indicate smaller reconstruction errors and therefore better reconstruction quality.

3.6 Model Variants Explored

Different inpainting configurations were evaluated during the development of MorphAI. LaMa was used as a pretrained baseline, while LGNet was used as the main inpainting model. A modified LGNet configuration was then developed by providing RGB images together with a binary facial mask and landmark heatmap. The main variants are summarized in Table 3.3, while their quantitative results are presented in Section 5.1.

Variant	**Description**
LaMa	Pretrained image inpainting model.
LGNet	Inpainting model used for facial reconstruction.
LGNet + 5-channel input	**Modified LGNet using RGB image, binary facial mask, and landmark heatmap for guided facial reconstruction.**

Table 3.3: Model variants explored during MorphAI development

CHAPTER 4

SYSTEM DESIGN

4.1 Requirement Specification

4.1.1 Functional Requirements

The proposed MorphAI system shall provide the following functional requirements:

- **Input Handling:** The system shall accept facial images from the CelebA-HQ dataset for training and evaluation.
- **Image Pre-processing:** The system shall resize input facial images to 256×256 pixels and prepare them for the reconstruction pipeline.
- **Facial Landmark Detection:** The system shall use MediaPipe Face Mesh to detect facial landmarks from the input image.
- **Mask Generation:** The system shall generate facial masks for selected regions using semantic facial regions and random rectangular masks.
- **Landmark Heatmap Generation:** The system shall convert the detected facial landmarks into a landmark heatmap to provide additional facial structure information.
- **Model Input Preparation:** The system shall combine the three RGB channels, one binary mask channel, and one landmark heatmap channel to create a five-channel input of size $5 \times 256 \times 256$.
- **Facial Reconstruction:** The system shall use the modified LGNet generator to reconstruct the masked facial regions and produce a 256×256 RGB facial image.
- **Identity Preservation:** The system shall use a pretrained facial recognition model to calculate identity loss and encourage the reconstructed image to preserve the identity of the input subject.
- **Model Training:** The system shall train the modified LGNet using a composite loss consisting of reconstruction, perceptual, style, identity, and adversarial objectives.

- **Evaluation:** The system shall evaluate reconstructed images using image-quality and identity-based metrics, including PSNR, SSIM, LPIPS, FID, RMSE, RMSE-log, and identity similarity.
- **Inference:** The system shall accept a facial image and generate a reconstructed facial image using the trained MorphAI model.
- **Output:** The system shall provide the reconstructed facial image while preserving the visible facial structure and attempting to maintain the identity of the subject.

4.1.2 Non-Functional Requirements

The MorphAI system should satisfy the following non-functional requirements:

- **Performance:** The system should use GPU acceleration to reduce model training and inference time.
- **Accuracy:** The system should generate realistic facial reconstructions while maintaining high structural and identity similarity.
- **Reliability:** The preprocessing and reconstruction pipeline should produce consistent results for valid facial images and different masking conditions.
- **Robustness:** The system should handle variations in facial expression, pose, lighting, and masked facial regions as effectively as possible.
- **Usability:** The system should provide a simple pipeline for preparing facial images, generating reconstructions, and evaluating the results.
- **Maintainability:** The system should use modular components for preprocessing, landmark detection, mask generation, model training, inference, and evaluation so that individual components can be modified or improved easily.
- **Scalability:** The framework should allow future experiments with larger datasets, different facial reconstruction models, and additional identity-preservation methods.
- **Resource Efficiency:** The system should be capable of training and testing within the available GPU and memory resources without requiring specialized 3D facial scanning hardware.

4.2 Use Case Diagram

The Use Case Diagram 4.1 illustrates the interaction between the user and the MorphAI Facial Appearance Transformation System.

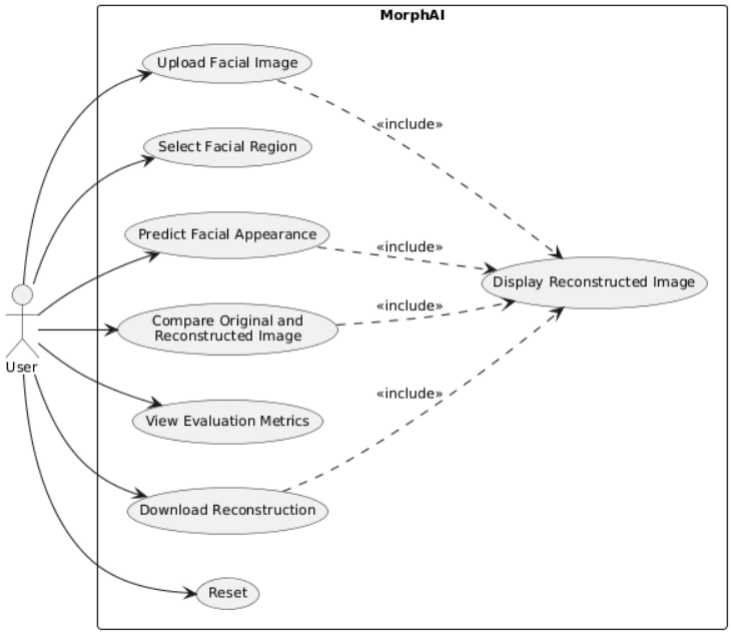


Figure 4.1: Use Case Diagram of MorphAI

The process begins when the user uploads a facial image and selects the facial region to be reconstructed. The user can then generate a facial appearance prediction, after which the reconstructed image is displayed. The user can compare the original and reconstructed images, view evaluation metrics, download the reconstruction, or reset the prediction.

4.3 Class Diagram for System

This class diagram 4.2 illustrates the structure and relationships among the major components of the MorphAI facial inpainting system.

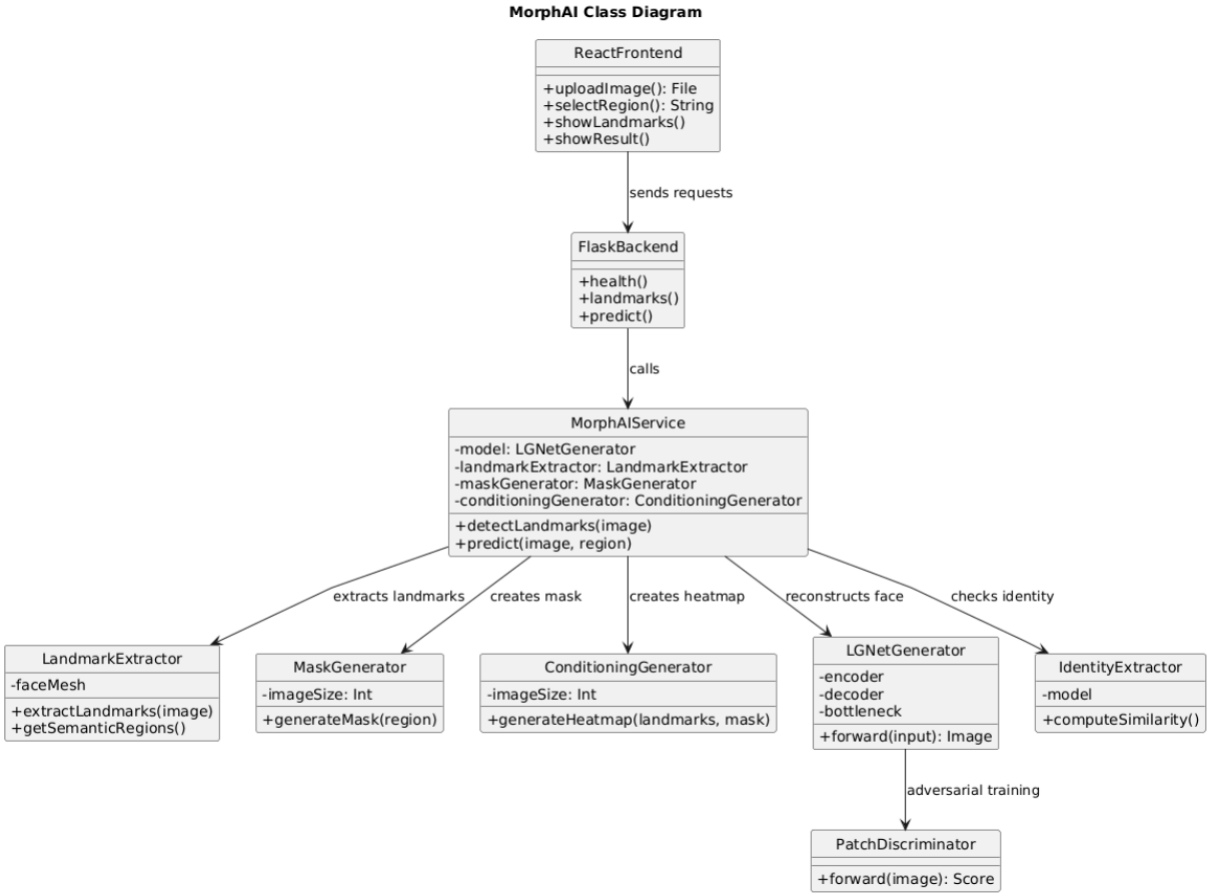


Figure 4.2: Class diagram of MorphAI system

The **ReactFrontend** serves as the entry point where the user uploads a facial image, selects the target facial region, and views the reconstructed result. It communicates with the **FlaskBackend**, which handles health checks, landmark extraction, and prediction requests. The **MorphAIService** coordinates the main processing components, including the **LandmarkExtractor**, **MaskGenerator**, **ConditioningGenerator**, **LGNetGenerator**, and **IdentityExtractor**. The **LandmarkExtractor** detects facial landmarks and semantic regions, while the **MaskGenerator** creates the mask for the selected facial region. The **ConditioningGenerator** uses the extracted landmarks and mask to generate the conditioning heatmap. These inputs are provided to the **LGNetGenerator**, which reconstructs the masked facial region using its encoder, decoder, and bottleneck components. The generated image is further evaluated through the **PatchDis-**

criminator during adversarial training and the **IdentityExtractor** to verify identity preservation. Each component has a well-defined responsibility, providing a structured flow from facial image input and region selection to reconstructed and identity-preserving facial output.

4.4 Sequence Diagram

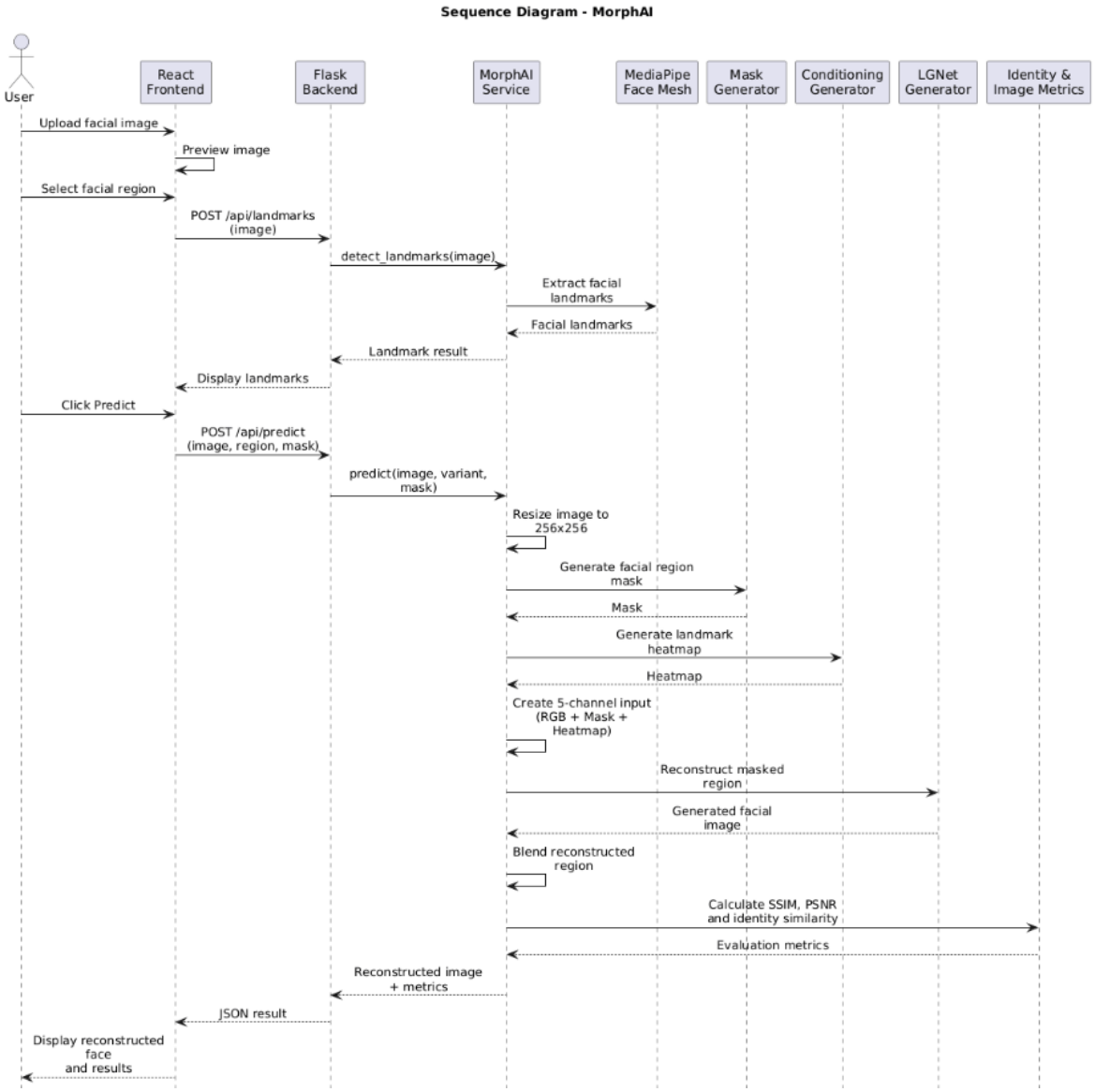


Figure 4.3: Sequence Diagram of MorphAI system

The sequence diagram 4.3 illustrates the interaction between system components during facial inpainting. The user uploads an image and selects a facial region through the React frontend. The Flask backend forwards the request to the MorphAIService, which extracts landmarks, generates a mask and heatmap, and forms the five-channel input for the LGNetGenerator. The generated facial image is then returned to the frontend for display, while the IdentityExtractor evaluates identity preservation.

The backend then sends the image to the MorphAIService, which generates the facial mask, landmark heatmap, and five-channel input for the LGNetGenerator. The reconstructed image is blended and evaluated using SSIM, PSNR, and identity similarity before being returned to the frontend and displayed to the user.

CHAPTER 5
RESULT AND DISCUSSION

5.1 Experiment Setup

All experiments for the proposed MorphAI system were conducted using a PyTorch-based deep learning environment with GPU acceleration. The model was trained using Automatic Mixed Precision (AMP) to improve computational efficiency and reduce GPU memory usage. Training was performed for five epochs on a dataset containing more than 16,800 facial images. Table 5.1 summarises the hardware and software environment used for model training and evaluation.

Component	**Specification**
GPU	NVIDIA GeForce RTX 4050 Laptop GPU (6 GB VRAM)
System RAM	16 GB
Operating System	Windows 11 Home
Python	3.11.9
PyTorch	2.6.0 + CUDA 12.4
Training Precision	Automatic Mixed Precision (AMP)
Training Epochs	5
Training Dataset	16,800 facial images

Table 5.1: Hardware and software environment used for MorphAI

5.2 Training Configuration

The proposed MorphAI model was trained using the modified LGNet generator with landmark-based structural conditioning. The input image was resized to 256×256 pixels and combined with a facial mask and landmark heatmap to form a five-channel input. The model was trained using GPU acceleration and Automatic Mixed Precision (AMP). The main training configuration is summarised in Table 5.2.

Training Setting	Value
Dataset	CelebA-HQ (16,800 original images)
Image Resolution	256×256
Model	Modified LGNet Generator
Input	5-channel (RGB + Mask + Heatmap)
Training Epochs	5
Training Precision	Automatic Mixed Precision (AMP)
Training Mask	Semantic facial-region masks
Inference Mask	Semantic masks
Evaluation	PSNR, SSIM, LPIPS, FID, RMSE-log, Identity Similarity

Table 5.2: Training configuration of the proposed MorphAI model

The model was trained for five epochs primarily due to the computational constraints of the available hardware. Training was performed on a single NVIDIA RTX 4050 Laptop GPU (6 GB VRAM), using Automatic Mixed Precision (AMP) to reduce GPU memory usage and improve computational efficiency within this resource budget. The training and validation losses decreased consistently throughout training, indicating stable convergence. The loss values obtained during training are presented in Table 5.3.

Epoch	Training Loss	Validation Loss	Loss Gap
1	0.3479	0.3640	0.0161
2	0.3124	0.3284	0.0160
3	0.3003	0.3167	0.0164
4	0.2847	0.2983	0.0136
5	0.2774	0.2950	0.0176

Table 5.3: Training and validation loss across five epochs

The training loss decreased from 0.3479 to 0.2774, while the validation loss decreased from 0.3640 to 0.2950. The relatively small and stable training–validation gap (ranging from 0.0136 to 0.0176) indicates stable training without a substantial increase in validation error, suggesting the model had not begun overfitting by the fifth epoch. Based on this consistent convergence behaviour and the available computational budget, the fifth-epoch checkpoint was selected as the final model for evaluation. Training was not extended beyond five epochs, so no claim is made regarding whether additional epochs would have produced further improvement or diminishing returns.

5.2.1 Ablation Consideration: Landmark Heatmap Conditioning

The contribution of landmark heatmap conditioning is discussed based on the model architecture and the overall reconstruction results reported in this chapter, rather than through a separate controlled experiment. A dedicated ablation study — training an identical LGNet variant without the landmark heatmap channel, under the same dataset, epoch budget, and evaluation protocol — was not conducted for this project due to GPU and time constraints (a single 6 GB VRAM GPU and a five-epoch training budget). The comparison against the LaMa baseline in Table 5.5, which does not use landmark conditioning at all, offers indirect evidence that the added structural guidance contributes to MorphAI's stronger PSNR, SSIM, and identity-similarity scores, but a same-architecture, single-variable ablation is left as future work.

5.2.2 Quantitative Evaluation

Semantic region-wise evaluation was performed on six facial regions using the same 2,800 held-out facial images. Each image was evaluated with respect to each of the six facial regions, resulting in $2,800 \times 6 = 16,800$ region-specific evaluation samples. As shown in Table 5.4, LGNet generally achieves higher PSNR, SSIM, identity similarity, and lower RMSE-log and FID than LaMa. LGNet also shows particularly strong performance for the mouth and eye regions. LPIPS is lower for LaMa in several regions, indicating that LGNet does not outperform the baseline on every metric. Overall, the results demonstrate strong structural and identity preservation by the proposed LGNet model.

Region	**Model**	**PSNR ↑**	**SSIM ↑**	**LPIPS ↓**	**ArcFace ID ↑**	**RMSE-log ↓**	**FID ↓**
Left Eyebrow	LGNet	42.36 dB	0.9932	0.0061	0.9880	0.0058	0.2439
Left Eyebrow	LaMa	42.10 dB	0.9812	0.0025	0.9726	0.0061	0.1581
Right Eyebrow	LGNet	42.27 dB	0.9931	0.0062	0.9869	0.0059	0.2422
Right Eyebrow	LaMa	42.03 dB	0.9811	0.0026	0.9690	0.0062	0.1549
Left Eye	LGNet	41.17 dB	0.9938	0.0051	0.9966	0.0067	0.2491
Left Eye	LaMa	39.97 dB	0.9819	0.0032	0.9880	0.0083	0.7603
Right Eye	LGNet	41.14 dB	0.9938	0.0051	0.9963	0.0067	0.2401
Right Eye	LaMa	40.05 dB	0.9819	0.0031	0.9869	0.0082	0.6960
Mouth	LGNet	39.87 dB	0.9947	0.0032	0.9886	0.0081	0.5940
Mouth	LaMa	34.59 dB	0.9734	0.0090	0.9497	0.0142	2.3734
Nose	LGNet	39.09 dB	0.9882	0.0092	0.9353	0.0080	0.7947
Nose	LaMa	37.53 dB	0.9754	0.0077	0.8660	0.0094	1.1818

Table 5.4: Semantic region-wise evaluation of MorphAI (LGNet) and LaMa on 2,800 samples per region

The final model was evaluated using pixel-level, perceptual, and identity metrics. The evaluation results were stored in JSON files, including `metrics.json`. A separate set of results was generated for comparison with the LaMa baseline.

Metric	**MorphAI (LGNet)**	**LaMa**
PSNR ↑	40.99 dB	39.38 dB
SSIM ↑	0.9928	0.9792
LPIPS ↓	0.0058	0.0047
Identity Similarity ↑	0.9819	0.9554
RMSE-log ↓	0.0069	0.0087
FID ↓	0.1398	0.2466

Table 5.5: Quantitative comparison between MorphAI and LaMa

MorphAI outperforms the LaMa baseline on PSNR, SSIM, identity similarity, RMSE-log, and FID, while LaMa achieves a slightly lower LPIPS score. Overall, these results indicate that the proposed landmark-conditioned LGNet provides strong reconstruction quality while maintaining facial identity.

5.2.3 Discussion of LPIPS and Identity Preservation

MorphAI does not outperform LaMa on every evaluation metric. LaMa achieves a slightly lower LPIPS score (0.0047 vs. 0.0058), indicating a lower perceptual feature distance according to the LPIPS metric. However, MorphAI achieves better PSNR, SSIM, ArcFace identity similarity, RMSE-log, and FID. This indicates that MorphAI provides stronger pixel-level reconstruction, structural consistency, and identity preservation, while LaMa has an advantage according to LPIPS.
This difference can be explained by what each metric is actually sensitive to. PSNR and SSIM measure pixel-level and structural similarity directly against the ground-truth image, and both reward MorphAI's landmark-conditioned reconstructions for closely matching the original facial structure. ArcFace identity similarity specifically rewards preservation of identity-discriminative features, which MorphAI's identity loss directly optimizes for during training. LPIPS, in contrast, measures similarity in a learned deep perceptual feature space that is more sensitive to fine-grained texture and stylistic detail than to precise structural or identity accuracy. It is likely that MorphAI's landmark and identity losses push the generator toward structurally and identity-accurate reconstructions, which can slightly reduce fine textural variation compared to LaMa's purely context-driven inpainting – a difference LPIPS is more sensitive to than PSNR or SSIM.

The region-wise results in Table 5.4 support this discussion. In the mouth region, for example, MorphAI achieves 39.87 dB PSNR, 0.9947 SSIM, and 0.9886 identity similarity, compared with LaMa's 34.59 dB, 0.9734, and 0.9497, respectively. Therefore, the results demonstrate a metric-specific trade-off rather than universal superiority of one model: MorphAI prioritizes structural and identity fidelity, while LaMa's context-based inpainting produces marginally lower LPIPS distance in some regions.

Region-wise evaluation was also performed for the eyes, eyebrows, nose, and mouth. These results are stored in the `lama_comparison` JSON files and provide additional analysis of performance across different facial regions.

5.3 Software Development Approach

The MorphAI project follows the Agile Software Development Life Cycle (SDLC), where development is carried out through iterative stages. Each iteration involves planning, implementation, testing, review, and refinement. This approach allows individual components to be developed and improved based on intermediate experimental results.

The development began with literature review, dataset preparation, image preprocessing, and facial landmark extraction. Subsequent iterations focused on facial mask generation, landmark heatmap conditioning, LGNet architecture modification, and ArcFace identity-preservation integration. The final stages focus on model training, evaluation, comparison, and refinement. Agile was selected because the project involves continuous experimentation with the dataset, model architecture, and loss functions, allowing changes to be made incrementally without affecting the entire system.

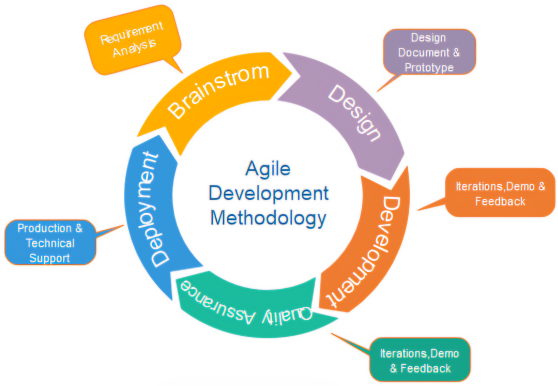


Figure 5.1: Agile Development Model

5.3.1 Agile Sprints

5.3.1.1 Sprint 1: Project Definition and Dataset Exploration

Duration: Initial Phase
Goal: Define the facial inpainting problem and identify suitable datasets for training and evaluation.

Key Activities:

- Reviewed existing facial inpainting and image restoration methods.
- Explored publicly available face datasets suitable for training and evaluation.
- Identified the need to preserve facial structure, texture, and identity during reconstruction.
- Defined the main evaluation criteria, including image quality and identity preservation.

Outcomes:

- Facial inpainting problem and project objectives were clearly defined.
- Suitable datasets and evaluation metrics were identified.
- Initial system requirements were established.

5.3.1.2 Sprint 2: Dataset Preparation and Mask Generation

Duration: Development Phase
Goal: Prepare the facial image dataset and generate masked training samples for different facial regions.

Key Activities:

- Collected and prepared face images for model training and evaluation.
- Generated masks for important facial regions such as the eyes, nose, mouth, and eyebrows.
- Applied facial alignment and image resizing to obtain consistent input dimensions.
- Prepared masked and corresponding ground-truth image pairs for training.

Outcomes:

- Training and evaluation data were prepared successfully.

- Region-specific facial masks were generated.
- The dataset was ready for LGNet training.

5.3.1.3 Sprint 3: LGNet Architecture and Training Pipeline

Duration: Development Phase
Goal: Design and implement the LGNet-based facial inpainting pipeline.

Key Activities:

- Designed the local and global context modules for facial feature reconstruction.
- Integrated facial masks and landmark information into the model input.
- Implemented the multi-objective loss combining L1, perceptual, style, identity, and landmark losses.
- Configured the PyTorch training pipeline with mixed-precision training.

Outcomes:

- LGNet training architecture was implemented.
- The complete training pipeline was successfully configured.
- Initial training runs confirmed that the model could reconstruct masked facial regions.

5.3.1.4 Sprint 4: Model Training and Optimization

Duration: Training Phase
Goal: Train LGNet and improve reconstruction quality while maintaining facial identity.

Key Activities:

- Trained the model using PyTorch automatic mixed precision across multiple epochs.
- Monitored training and validation losses to evaluate convergence.
- Adjusted the training configuration based on observed model performance.
- Saved model checkpoints and selected the best-performing checkpoint for evaluation.

Outcomes:

- Training loss decreased consistently throughout training.
- The model reached stable convergence without significant overfitting.
- The fifth-epoch checkpoint was selected as the final model checkpoint.

5.3.1.5 Sprint 5: Boundary Blending and Identity Preservation

Duration: Refinement Phase
Goal: Improve the visual quality of reconstructed regions and preserve the identity of the input face.

Key Activities:

- Implemented alpha feathering to reduce visible boundaries around reconstructed regions.
- Applied Poisson blending to improve colour and illumination consistency.
- Used facial identity embeddings to evaluate identity preservation.
- Verified facial landmark consistency between reconstructed and original images.

Outcomes:

- Boundary artifacts were reduced after post-processing.
- Reconstructed regions achieved better visual and structural consistency.
- Identity preservation was incorporated into the final evaluation process.

5.3.1.6 Sprint 6: Evaluation and Comparative Analysis

Duration: Final Phase
Goal: Evaluate the final model and compare its performance with the LaMa baseline.

Key Activities:

- Evaluated the final model on the held-out test samples.
- Computed PSNR, SSIM, LPIPS, $\text{RMSE}_{\log}$, FID, and ArcFace identity scores.
- Analysed performance separately for different facial regions.

- Compared LGNet results with the LaMa baseline using the stored evaluation metrics.

Outcomes:

- Final quantitative results were obtained for the proposed model.
- LGNet showed strong reconstruction quality and identity preservation.
- The final system and evaluation results were prepared for demonstration and reporting.

5.4 Model Deployment

This section describes the deployment of the MorphAI system, including the frontend interface, backend services, trained LGNet model, and the complete inference pipeline.

5.4.1 Frontend Development

The frontend was developed using React to provide a simple interface for uploading a facial image and selecting the region to be reconstructed. The main components include:

- Facial image upload with image preview.
- Selection of facial regions such as the eyes, nose, eyebrows, and mouth.
- Display of the reconstructed facial image.
- Before-and-after comparison of the input and generated images.
- Status messages for image processing and inference.

Screenshots of the user interface are provided in Appendices C.1.

5.4.2 Backend Development

The backend was implemented using Flask to handle image processing and model inference. Its main responsibilities include:

- Receiving the uploaded facial image and selected facial region.
- Extracting facial landmarks using the landmark extraction module.

- Generating a region-specific mask for the selected facial area.
- Creating the facial landmark heatmap required by the model.
- Combining the RGB image, mask, and heatmap to form the five-channel model input.
- Running inference using the trained LGNet generator.
- Applying boundary blending to reduce visible seams around the reconstructed region.
- Returning the final reconstructed image to the frontend.

5.4.3 Model Integration

The trained LGNet generator was integrated into the backend for real-time facial inpainting. The complete inference process follows these steps:

- The user uploads an image and selects a facial region.
- The backend extracts facial landmarks and generates the corresponding mask.
- The RGB image, mask, and landmark heatmap are combined into the required five-channel input.
- LGNet generates the reconstructed facial region.
- Boundary blending is applied to obtain a visually seamless result.
- The reconstructed image is returned to the React frontend for display.

The system can additionally use the identity extraction module to compare the reconstructed face with the original face using facial embeddings. This helps verify whether the reconstructed result preserves the identity of the input subject.
Overall, the deployment integrates the React frontend, Flask backend, facial preprocessing modules, and LGNet generator into a single end-to-end facial inpainting system.

CHAPTER 6
CONCLUSION

6.1 Conclusion

This project successfully developed MorphAI, an identity-preserving facial inpainting system based on the LGNet architecture. The proposed system reconstructs masked facial regions while maintaining facial structure, local textures, and the identity of the input subject. The pipeline combines facial landmark extraction, region-specific mask generation, landmark heatmaps, and a five-channel input representation for the LGNet generator.

The model was trained for five epochs using mixed-precision training on a dataset containing more than 16,800 facial images. The training loss decreased consistently, with the fifth epoch selected as the final convergence checkpoint. Evaluation was performed on 16,800 test samples across different facial regions and compared with the LaMa baseline.

The proposed LGNet achieved an overall PSNR of 40.99 dB, SSIM of 0.9928, ArcFace identity score of 0.9819, RMSE_{log} of 0.0069, and FID of 0.1398. Compared with LaMa, LGNet provided better structural and identity preservation, particularly for challenging regions such as the mouth and nose. These results demonstrate that the proposed approach can effectively restore facial regions while maintaining the visual characteristics and identity of the original face.

6.2 Limitations

Despite the promising results, the system has several limitations:

- The system is primarily designed for 2D facial image inpainting and does not currently reconstruct or represent faces in 3D.
- The current implementation is not designed as a live or real-time video inpainting system and is mainly intended for single-image reconstruction.
- The quality of reconstruction depends on the accuracy of facial landmark detection and the generated region mask.
- Performance may decrease for heavily corrupted regions or facial structures with limited surrounding visual information.

- The model requires GPU resources for efficient inference and training, which may limit deployment on low-resource devices.
- Although ArcFace similarity is used to evaluate identity preservation, the identity score does not guarantee perfect preservation in every reconstructed image.

6.3 Future Works

Based on the findings of this project, several improvements can be explored in future work:

- **3D Facial Reconstruction:** Extend MorphAI from 2D image inpainting to 3D facial reconstruction to model facial geometry and provide more realistic results from different viewpoints.
- **Real-Time Video Inpainting:** Extend the system to process video streams and reconstruct masked facial regions continuously while maintaining temporal consistency between frames.
- **Dataset Expansion:** Train the model on larger and more diverse facial datasets containing different poses, lighting conditions, expressions, and degrees of facial occlusion.
- **Improved Identity Preservation:** Explore stronger identity-aware objectives and modern face-recognition embeddings to further improve preservation of facial identity.
- **Advanced Generative Models:** Investigate diffusion-based and other modern generative architectures to improve reconstruction quality for complex facial regions.
- **Mobile and Edge Deployment:** Optimize the model through techniques such as quantization, pruning, or knowledge distillation to enable efficient deployment on resource-constrained devices.
- **Interactive Reconstruction:** Develop a more flexible interface that allows users to select, adjust, and preview different facial regions before generating the final reconstruction.

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APPENDIX A

Raw Dataset

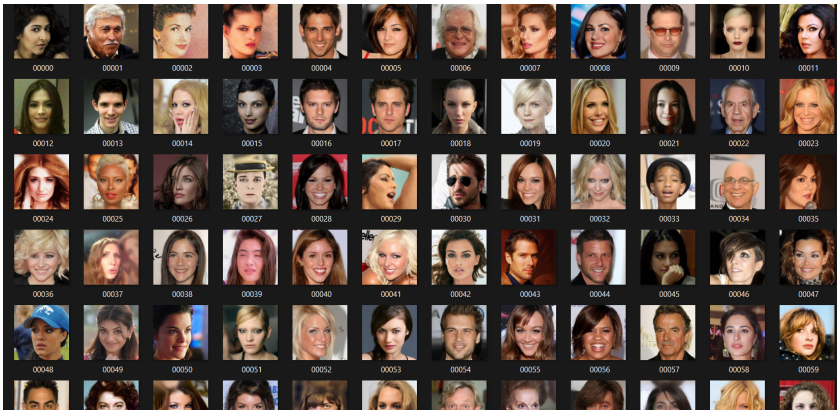


Figure A.1: Sample raw dataset

APPENDIX B

Data Processed

B.1 Data1



Figure B.1: Main interface

B.2 Data1 Landmark

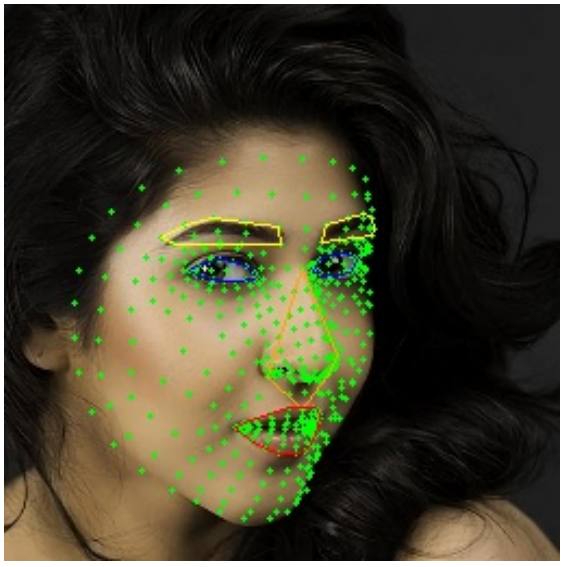


Figure B.2: Input Image

B.3 Semantic Masks



Figure B.3: Semantic mask for the left eye region

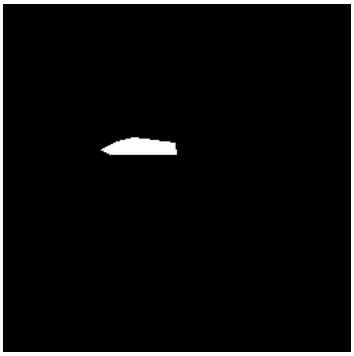


Figure B.4: Semantic mask for the left eyebrow region

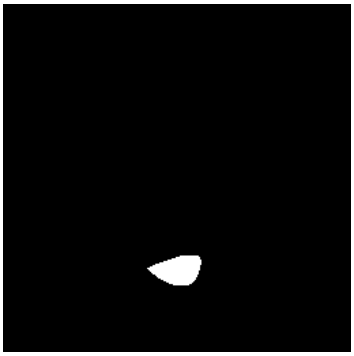


Figure B.5: Semantic mask for the mouth region

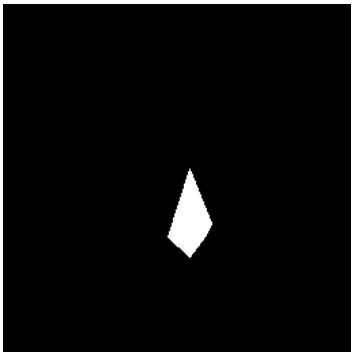


Figure B.6: Semantic mask for the nose region

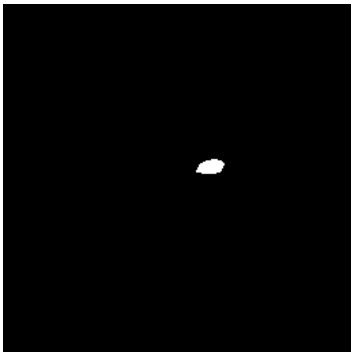


Figure B.7: Semantic mask for the right eye region

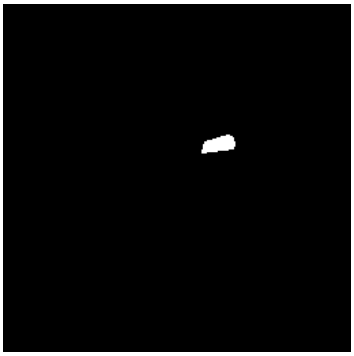


Figure B.8: Semantic mask for the right eyebrow region

B.4 Heatmaps

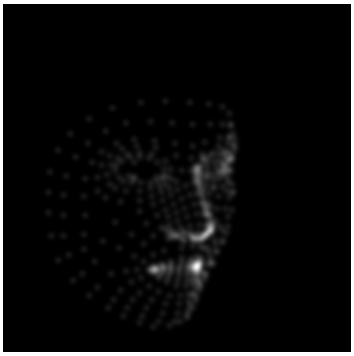


Figure B.9: Landmark heatmap for the left eye region

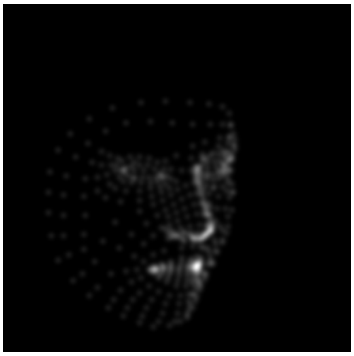


Figure B.10: Landmark heatmap for the left eyebrow region

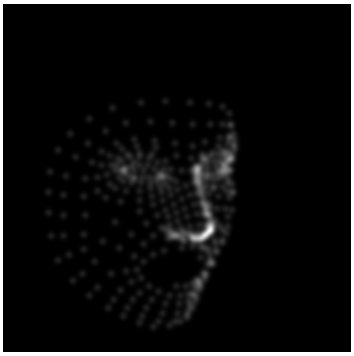


Figure B.11: Landmark heatmap for the mouth region

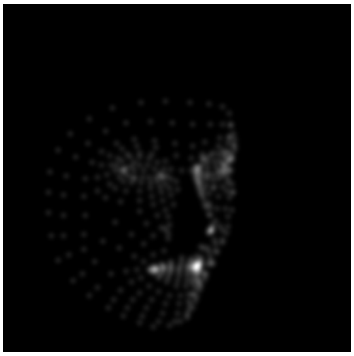


Figure B.12: Landmark heatmap for the nose region

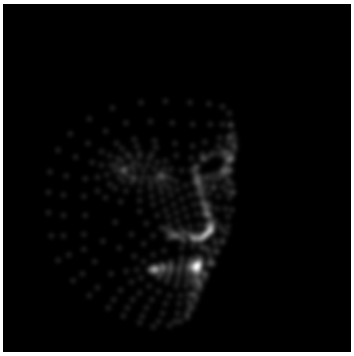


Figure B.13: Landmark heatmap for the right eye region

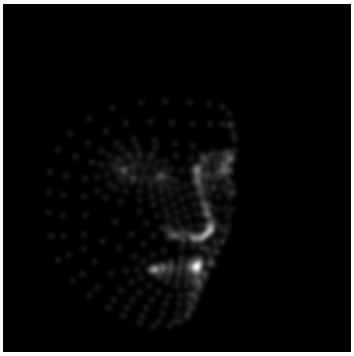


Figure B.14: Landmark heatmap for the right eyebrow region

APPENDIX C

Image Pre-processing Pipeline Details

C.1 Original Input (256*256)



Figure C.1: Original Input (256*256)

C.2 Landmarks Detection



Figure C.2: Landmarks Detection

C.3 Masked Input



Figure C.3: Masked Input

C.4 Binary Mask

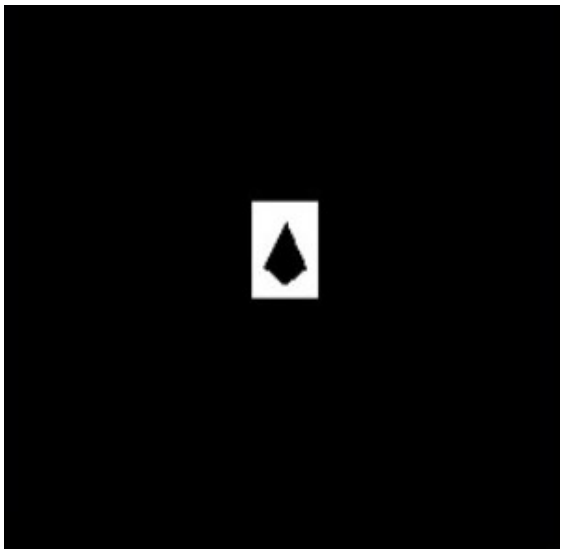


Figure C.4: Binary Mask

C.5 Landmark Heatmap

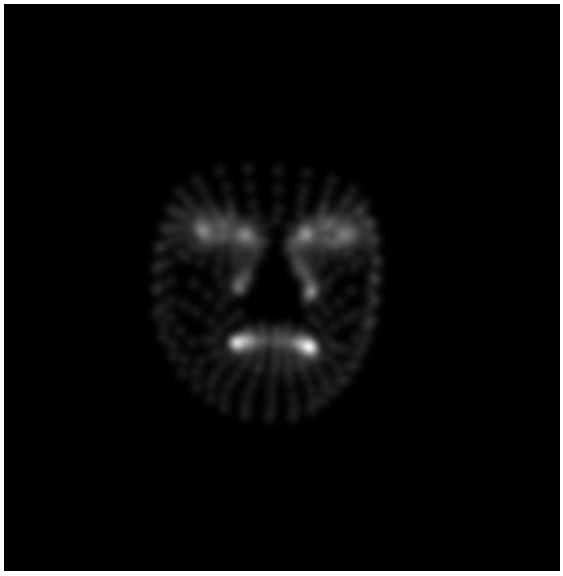


Figure C.5: Landmark Heatmap

C.6 Raw Generator Output



Figure C.6: Raw Generator Output

C.7 Reconstructed Output



Figure C.7: Reconstructed Output

APPENDIX D

User Interface Screenshots

D.1 Main Interface

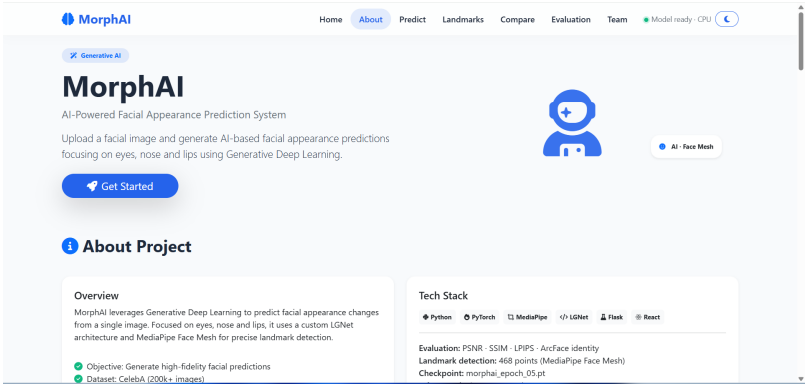


Figure D.1: Main interface

D.2 Input Image

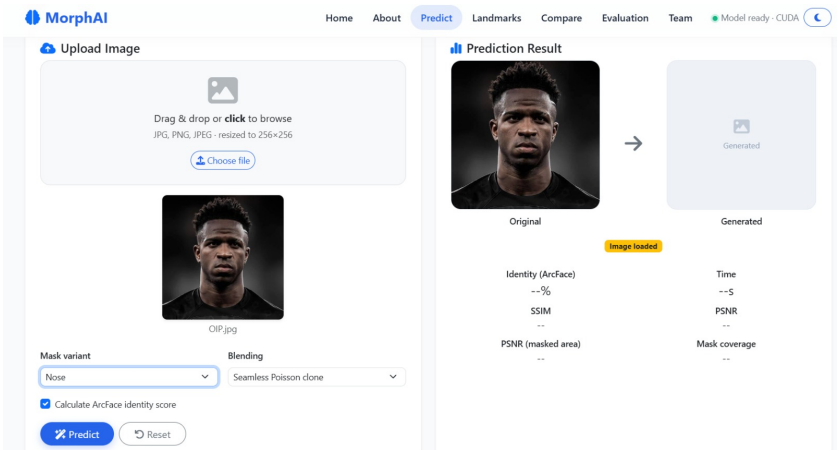


Figure D.2: Input Image

D.3 Facial Landmark

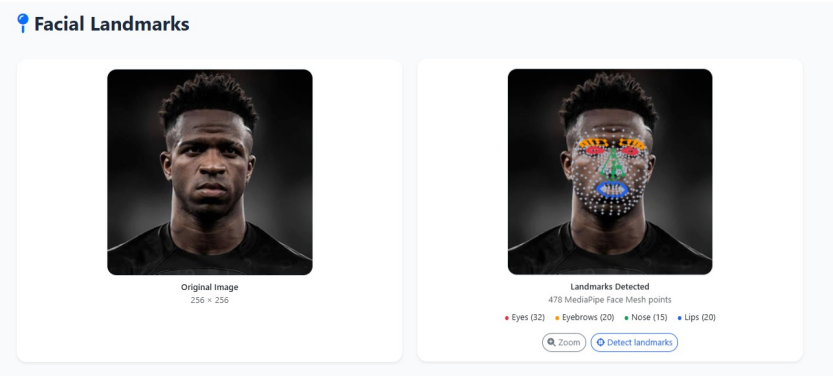


Figure D.3: Facial Landmark

D.4 Inference Pipeline

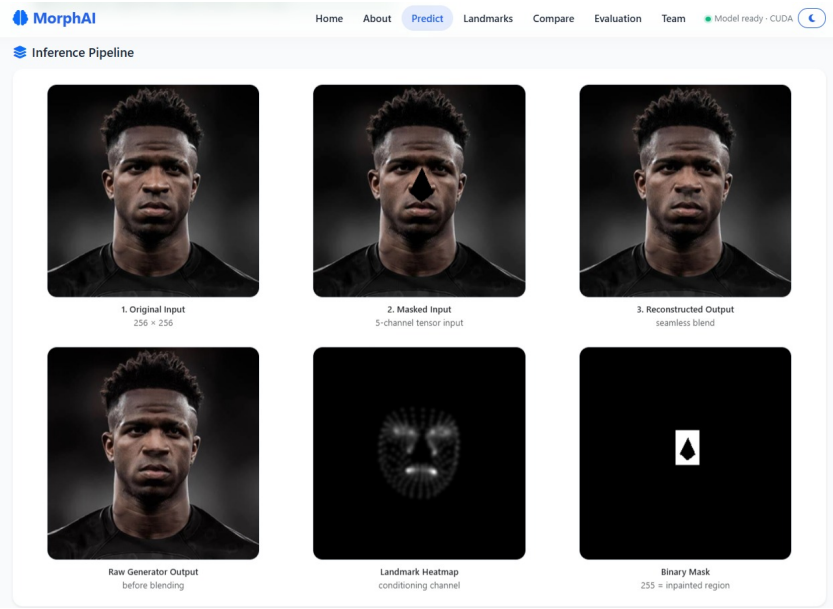


Figure D.4: Inference Pipeline

D.5 Predicted Output

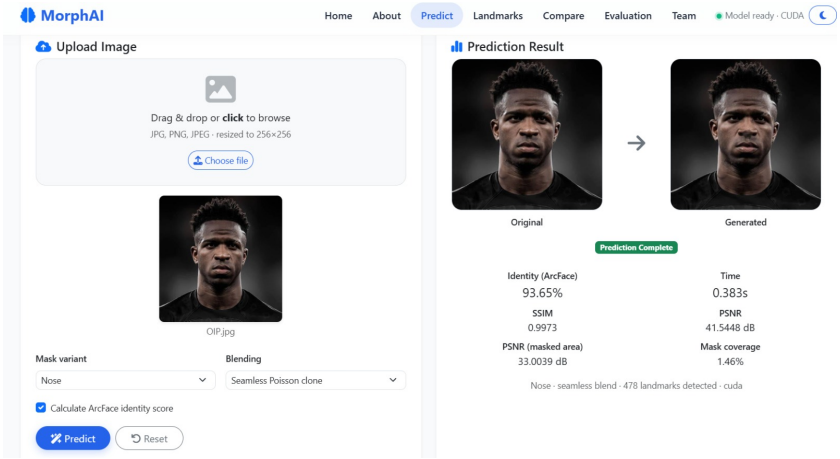


Figure D.5: Predicted Output